

Labwork 7: Grayscale Stretch

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1 Method

I make a low-contrast input from the Lab 3 RGB image by dividing each channel by 2 and adding 50. The GPU first converts it to grayscale. Each block then finds its minimum and maximum using shared-memory reduction. I use the last reduction loop from the slide: the stride starts at half of the block and is divided by 2 at each step. I repeat the reduction on the block results until only one minimum and one maximum remain. Threads outside the image use 255 for minimum and 0 for maximum, so they do not change the result.

The last kernel calculates $(\text{gray} - \text{min}) * 255 // (\text{max} - \text{min})$. For a constant image, I write 0 to avoid division by zero.

2 Results

I test three 2D block sizes. I compile the kernels before timing. Times include the grayscale conversion, all reduction passes and their allocations, stretch, and data transfers. Each configuration is measured once.

Block size: (8, 8)
Min: 50 Max: 177
Total time: 3.853 ms
Block size: (16, 16)
Min: 50 Max: 177
Total time: 2.121 ms
Block size: (32, 32)
Min: 50 Max: 177
Total time: 1.678 ms

The input grayscale range is 50 to 177; the output range is 0 to 255. All three configurations give the same result. The 32 by 32 block has the lowest time in this run.



Grayscale input and stretched result.